

Provenance-Constrained Selective Language Style Planning from Sparse Evidence

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Abstract

Memorial voice systems pose a language-generation problem before they pose a speech-synthesis problem: a system may have only a few historical messages and conflicting recollections, yet fluent output can be mistaken for words or beliefs that a person actually expressed. We formulate *provenance-constrained language style planning*, in which a planner must distinguish supported content from observable linguistic style, attribute every accepted claim to consent-bearing evidence, preserve source conflicts, and abstain when support is insufficient. We introduce a typed multi-source evidence representation, a sparse style encoder, evidence-level attribution heads, and a selective decision rule. We also specify a speaker-disjoint living-speaker proxy protocol that varies textual evidence budgets and evaluates citation quality, conflict handling, calibration, and style distance. A downstream realizer is specified only as an approved-text interface; acoustic validation is outside this language paper. On the public PersonaChat role-instance proxy, a frozen MiniLM encoder reaches $69.17\% \pm 0.58$ accuracy over three seeds. We additionally construct 18,000 compositional episodes from independently collected SNLI relation labels and PersonaChat style labels. On its untouched test split, the planner obtains $90.39\% \pm 0.33$ citation precision and $74.03\% \pm 0.87$ recall, but the development risk constraint does not transfer: positive-case coverage is only $3.73\% \pm 4.24$, with $23.62\% \pm 3.17$ false acceptance among accepted cases. We report this failed selective-risk hypothesis rather than claim a safety guarantee. The benchmark is a controlled composition, not evidence about natural memorial speech or living-speaker authorization. An auxiliary DailyDialog audit shows lexical support coverage increasing from 28.81% to 55.74% as history grows from one to five turns, while top-1 source precision falls from 100.00% to 70.84%; these

weak overlap labels are a cross-domain structural audit, not planner or entailment results.

1 Introduction

A person’s historical record is rarely balanced. There may be thousands of messages but only a second of clean speech, or a short recording but sparse text. Family members may remember a habitual phrase, tone, or event differently. A generative system can nevertheless produce a coherent sentence and a plausible voice. Fluency creates a dangerous ambiguity: resemblance in style may be interpreted as evidence that the person said, believed, or would have said the generated content.

Existing zero-shot text-to-speech (TTS) systems study reconstruction from short references (Chen et al., 2025; Peng et al., 2024). Work on personality from language predicts aggregate trait labels (Mairesse et al., 2007; Schwartz et al., 2013), and grounded generation studies citation correctness (Gao et al., 2023). These threads do not by themselves define what a memorial system may say when records are sparse, heterogeneous, and disputed. Improving speaker similarity without constraining content can make unsupported output more persuasive.

We therefore separate three objects that are often conflated: (i) textual claims supported by first-party or documented evidence, (ii) observable linguistic-style statistics estimated from first-party text, and (iii) acoustic controls used to realize an already approved utterance. Big Five (OCEAN) values are treated as uncertain, provenance-bearing controls, not recovered psychological truth. A relative’s recollection may inform a disputed record or a control suggestion, but it is never converted into a verbatim statement by the represented person.

The resulting NLP task is useful without speech: given a candidate utterance and a small typed evidence set, select supporting evidence, identify conflict, measure style compatibility, and either accept

with citations or abstain. Any downstream speech layer receives only accepted text and is evaluated in a separate study; it is not part of the language claims here.

This work makes four contributions. First, it defines a typed evidence and claim contract that preserves source, consent, time, evidence type, and stance. Second, it introduces a selective planner with per-evidence attribution, explicit conflict scoring, observable style estimation, and fail-closed decision constraints. Third, it provides a large independently labelled compositional benchmark with official train/development/test separation. Fourth, it evaluates equal-budget contextual retrieval and selective RAG baselines, calibration, and errors across evidence budgets. The planner exposes an approved-text interface for downstream realizers, but this paper makes no acoustic contribution. We do not claim to reconstruct a personality, infer what a deceased person would think, or improve bereavement outcomes.

2 Task Definition

2.1 Typed evidence

For a target speaker s , the authorized evidence set is $E_s = \{e_i\}_{i=1}^n$. Each item is a tuple

$$e_i = (x_i, t_i, r_i, \tau_i, c_i, q_i, u_i), \quad (1)$$

where x_i is text, t_i is evidence type, r_i is source identity, τ_i is time period, c_i is consent status, q_i is stance (support, contradiction, or uncertainty), and u_i is source confidence. Types distinguish direct messages and recorded transcripts from family recollections, documented facts, and style observations. The representation does not imply that all sources have equal authority.

The most conservative executable policy accepts only a normalized exact match to a consented first-party message or transcript. This provides a non-neural lower bound with maximal precision but low coverage. Learned planning is allowed to exceed this coverage only when claim annotations and citation evaluation show that support is retained.

2.2 Outputs and fail-closed semantics

Given candidate y , the planner outputs support probability p_s , style-match probability p_y , conflict probability p_c , a selectively trained score p_{sel} when all joint labels are available, and attribution scores a_i for every evidence item. With thresholds selected only on development speakers, synthesis is

permitted iff

$$p_a \geq \theta_a \wedge p_s \geq \theta_s \wedge p_c < \theta_c \wedge \exists i : a_i \geq \theta_e. \quad (2)$$

Otherwise the planner returns ABSTAIN and machine-readable reasons. Here $p_a = p_{sel} \max_i a_i$ for jointly labelled episodes. For style-only or legacy checkpoints, p_{sel} is replaced by the auditable product $p_s p_y (1 - p_c)$. Independent support, conflict, and citation gates remain mandatory, so a learned score cannot bypass a failed evidence gate. A fluent answer without an attributed evidence item cannot pass.

Accepted output remains synthetic. Citation means that specified evidence supports a claim under the annotation policy; it does not mean that the target speaker authored the generated sentence. Interfaces must preserve this distinction in disclosure and audit records.

3 Method

3.1 Sparse observable style

We avoid treating Big Five prediction as a direct proxy for idiolect. Instead, the auditable style vector contains message and sentence length, length variance, type–token ratio, function and discourse markers, punctuation, questions and exclamations, politeness and uncertainty markers, first-person rate, code switching, emoji rate, and lexical preferences. Only consented, non-disputed first-party records contribute to this vector.

For the executable baseline, token unigrams and local bigrams are mapped to a signed hashing vector $v(x) \in \mathbb{R}^d$ and normalized. A small feed-forward encoder g produces $h_x = g(v(x))$. Evidence embeddings are mean pooled, $h_E = n^{-1} \sum_i h_{x_i}$, and paired with candidate embedding h_y as

$$h_{E,y} = [h_E; h_y; |h_E - h_y|; h_E \odot h_y]. \quad (3)$$

The hashed bag-of-ngrams encoder is deliberately modest: it establishes a reproducible CPU baseline and makes failures inspectable. Our contextual model uses a frozen MiniLM encoder followed by a trainable projection and the same decision heads. Encoder revision, sequence limit, and initialization seed are stored in the checkpoint; frozen foundation weights are reloaded by revision rather than duplicated. Private memorial data are not sent to a remote inference API.

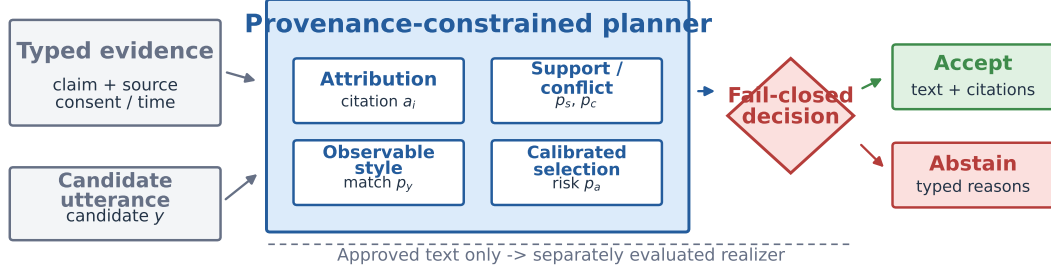


Figure 1: Language-only method overview. Typed evidence and a candidate utterance are encoded separately; attribution, support, style, and conflict heads feed a calibrated fail-closed decision. Downstream realization is an interface, not a result claimed by this paper.

Held-out authorship is used as a proxy task: evidence messages precede the candidate chronologically, and negatives come only from another speaker in the same split. This tests whether style features identify a writer rather than memorizing target-speaker IDs. Topic leakage is measured through topic-matched negative subsets, following the concern that author representations may encode content rather than style (Wegmann et al., 2022).

3.2 Support, attribution, and conflict

Rather than mean-pooling every source equally, the contextual model computes candidate-conditioned evidence weights

$$\alpha_i = \text{softmax}_i(h_i^\top h_y / \sqrt{d}), \quad h_E = \sum_i \alpha_i h_i. \quad (4)$$

This attention is not itself a citation: the citation head below receives direct supervision and is evaluated against evidence IDs. An ablation restores mean pooling.

The frozen observable-style features from the previous subsection are computed separately for evidence and candidate. Their vectors and absolute difference are projected to h_o and appended to $h_{E,y}$. Removing h_o tests whether the neural encoder, rather than explicit linguistic features, accounts for a gain. Separate heads map $[h_{E,y}; h_o]$ to support, style, conflict, and selective logits. The selective head is trained only when all joint labels are present; missing axes remain masked rather than imputed. Evidence attribution cannot be recovered from an aggregate head, so each item is also paired with the candidate:

$$h_{i,y} = [h_i; h_y; |h_i - h_y|; h_i \odot h_y], \quad (5)$$

$$a_i = \sigma(w_e^\top h_{i,y} + b_e).$$

Acceptance combines the calibrated selective score with the maximum evidence-attribution score. The independent component gates in Equation 2 keep the decision inspectable and fail closed.

Support and conflict require claim-level annotation. Crucially, a held-out message by the same writer supplies a style label but not a support label: the speaker’s earlier messages need not entail the later message. Missing task labels are masked rather than silently set to zero. This prevents an easy but invalid training shortcut.

For available labels, the objective is

$$\mathcal{L} = \mathcal{L}_{\text{style}} + m_s \mathcal{L}_{\text{support}} + m_c \mathcal{L}_{\text{conflict}} + m_e \mathcal{L}_{\text{cite}} + m_{\text{joint}} \mathcal{L}_{\text{sel}}. \quad (6)$$

where each component is binary cross entropy and m indicates that the corresponding annotation exists. No cross-dataset pseudo-label is introduced for a missing axis.

3.3 Calibration and abstention

Selective prediction is central rather than an auxiliary metric (Xin et al., 2021; Kamath et al., 2020). Temperature or isotonic calibration, decision thresholds, and the citation threshold are fit on development speakers only. We report expected calibration error (ECE), coverage, selective risk, false acceptance, and false abstention. A single accuracy at $\theta = 0.5$ is insufficient because the intended operating point prioritizes avoiding unsupported accepted speech.

Conflicting sources are not averaged into a confident pseudo-consensus. Any gold contradiction must either be surfaced or trigger abstention. We evaluate conflict precision and recall, source-pair breakdowns, and a conflict preservation score.

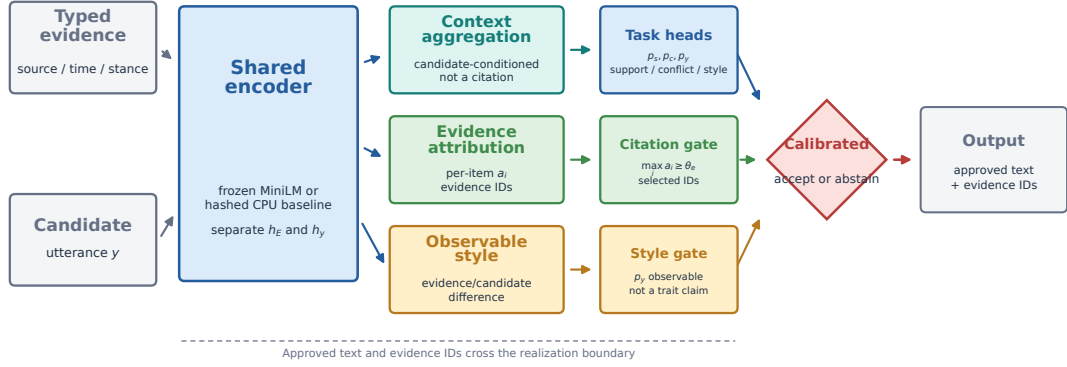


Figure 2: Mechanism detail. Candidate-conditioned attention forms an aggregate context but is not treated as provenance. A separately supervised attribution head predicts evidence IDs, while support, conflict, and observable-style signals remain inspectable before calibrated selection.

Recollection-only evidence cannot establish a first-party quotation even if sources agree.

3.4 Downstream realization boundary

The planner emits an auditable approved-text record containing the selected evidence IDs, decision reasons, and disclosure status. A downstream speech realizer may consume this record, but it must not bypass the planner or turn style evidence into a first-person factual claim. Voice cloning, acoustic controls, reference-duration curves, and listening studies require a separate evaluation; no acoustic result is used to support the language claims in this paper.

4 Public Proxy and Independent Labels

The preferred dataset contains living speakers who explicitly consent to research use. For public reproducibility, an openly licensed PersonaChat or ConvAI2-style export (Zhang et al., 2018) may be used as a PUBLIC-RESEARCH, text-only proxy; it grants no permission to clone a participant’s voice and is not a memorial source. FEVER- and SciFact-style evidence data (Thorne et al., 2018; Wadden et al., 2020) can evaluate support and conflict heads but cannot establish personal style. Each record retains stable speaker and conversation IDs, timestamp, consent scope, license identifier, and source. Private text is stored locally; release manifests contain hashes and non-sensitive metadata. The exact composite protocol and conversion script are documented in docs/research/PUBLIC_DATASETS.md.

4.1 Independent compositional benchmark

The principal controlled benchmark combines labels that were collected independently of our system. SNLI supplies human ENTAILMENT, NEUTRAL, and CONTRADICTION relations (Bowman et al., 2015); PersonaChat supplies a separate role-instance style label. We preserve SNLI’s official splits, sample each relation equally, and pair every sampled relation with one positive and one negative style episode. This produces 12,000 training, 3,000 development, and 3,000 test episodes.

Each episode contains the annotated premise, three length-matched SNLI premise distractors, and only the PersonaChat evidence supplied with that episode. The gold premise position rotates deterministically, preventing a positional shortcut. A citation identifies the premise on which the annotated NLI relation was judged; for neutral or contradictory hypotheses it does not mean that the premise supports the hypothesis as true. All policies receive this same episode-local evidence budget. SNLI is CC BY-SA 4.0 and PersonaChat is CC BY 4.0; neither requires an approval workflow. Source hashes, construction seed, label mapping, and the final manifest fingerprint are frozen.

This composition supplies independent evidence, relation, style, and citation targets at useful scale, but it is deliberately not described as naturally generated personal speech. A single SNLI premise is not a consent record, and a PersonaChat role is not a stable living-speaker identity.

Speakers, not messages, are assigned to train, validation, and test. The split is produced by a seeded hash ranking and checked so no target or negative candidate speaker crosses a split. Retrieval

indexes, prompts, calibration, and model selection are also split-scoped. With at least six speakers, each holdout receives at least two speakers to support within-split negatives.

For each speaker, the chronologically last eligible message is a candidate and earlier messages form evidence budgets $B \in \{1, 2, 5, 10, 25, \text{all}\}$. Positive style pairs use the same speaker; negative pairs use another speaker within the same split. In addition to the cyclic negative baseline, we build a topic-matched condition by selecting a different role instance with maximal token-Jaccard overlap using only split-local candidate text. Full planner episodes additionally require atomic claim support, contradiction, and evidence-attribution labels. Annotators see source type and time but not model predictions. Ambiguous cases retain an UNCERTAIN label rather than forced consensus.

4.2 Annotation protocol

Candidate utterances are segmented into minimal independently checkable claims. Pure discourse markers receive a style-only tag. For each remaining claim, two annotators independently select zero or more evidence IDs and one relation: SUPPORTED, CONTRADICTED, BOTH, UNCERTAIN, or NOT-ADDRESSED. BOTH is retained when authorized sources support incompatible values; it is not converted to a majority label. Annotators also mark exact quotation, close paraphrase, or novel wording.

The guide distinguishes statement attribution from world truth. A direct message supports the claim that its author wrote the message, but not automatically every proposition inside it. A recollection can support “source r recalls phrase p ,” but cannot support “speaker s said p ” without first-party corroboration. Time-sensitive claims must cite evidence from the relevant period.

Agreement is reported separately for claim boundaries, evidence selection, and relations. Evidence selection uses set F1 and relation labels use nominal Krippendorff’s alpha. Development disagreements refine the guide; test disagreements are adjudicated under the frozen guide. We report both pre-adjudication agreement and adjudicated scores. Annotators are never asked to infer a clinical or psychological state.

4.3 Data quality and exclusion gates

Records are excluded before splitting when authorization or license fields are missing, the speaker

ID cannot be stabilized, text is machine generated, or a timestamp required for chronological construction is unavailable. Duplicate and near-duplicate messages are grouped before splitting and cannot occur as both evidence and held-out targets. Forwarded or quoted spans are marked so they do not become evidence of the account owner’s style.

Language and script are recorded per message. Code-switched messages remain but are analyzed separately. Minimum speaker and message counts are declared before training. If a language lacks the predeclared number of test speakers, its results are descriptive and are not pooled into the principal endpoint.

5 Experimental Protocol

5.1 Language baselines and ablations

The frozen baselines are: (1) neutral generic text with no target style; (2) consented exact quotation; (3) retrieval-only nearest quotation; (4) retrieval-augmented generation (Lewis et al., 2020) with citations and no style component; and (5) the auditable hashed style planner. Equal-budget contextual baselines score only the evidence supplied in each episode: MiniLM and MPNet top-1 retrieval, plus a top-2 selective RAG-style attribution policy whose score threshold is fit on development data at a 90% citation-precision floor. Encoder names and exact revisions are stored in the artifact. If a larger language model is evaluated, its prompt, decoding parameters, model revision, and any remote data retention must be reported.

Ablations remove provenance constraints, attribution supervision, conflict handling, abstention, observable style features, and chronological evidence selection one at a time. We also compare first-party-only evidence against recollection-enhanced retrieval. The latter must never improve coverage by mislabeling recalled wording as a quotation.

5.2 Predeclared hypotheses

H1 concerns coverage and risk. Exact quotation should have the lowest unsupported-acceptance risk but limited coverage; the selective planner should improve coverage at a fixed citation-precision floor. The claim fails if coverage rises by accepting recollection-only paraphrases as first-party speech.

H2 concerns sparse style. Held-out authorship and observable-style distance should improve with

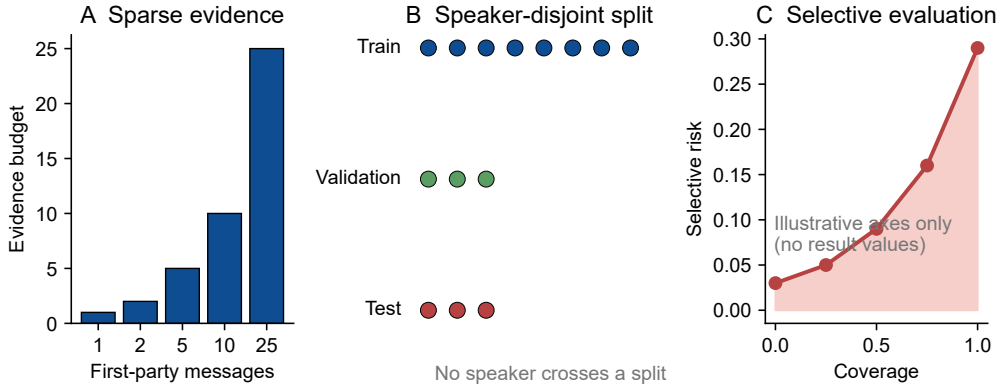


Figure 3: Evaluation design. Panel C illustrates the axes to be reported and contains no experimental result. Final curves are generated only from a measured evaluation artifact.

evidence budget, with diminishing gains. A gain that vanishes under topic-matched negatives counts as topic exploitation. H3 concerns conflict: explicit conflict supervision should improve contradiction recall without an unacceptable coverage loss on agreed claims.

Each hypothesis has one principal contrast; other analyses are secondary. Hyperparameters and thresholds are frozen after development. We report all seeds and never select a seed by test performance. If a hypothesis fails, its architectural claim is removed rather than rewritten around a favorable secondary metric.

5.3 Language metrics

Citation precision and recall operate on evidence IDs; supported-claim coverage operates on atomic claims. Factual precision is reported at the claim level, in the spirit of fine-grained evaluation (Min et al., 2023), but against the authorized evidence set rather than an open web corpus. Conflict precision, recall, and F1 test whether disputed material is surfaced.

Style is measured with held-out authorship accuracy, retrieval rank, and normalized distance over the frozen observable feature schema. We report both cyclic and token-Jaccard topic-matched negative conditions. Exact and near-verbatim overlap detect privacy memorization. Calibration uses ECE and reliability plots (Guo et al., 2017); selective behavior uses risk-coverage and error rates at a preregistered low-risk operating point.

All confidence intervals bootstrap speakers, not messages. Paired comparisons use speaker-level aggregates, with effect sizes and confidence intervals. Message-level tests would inflate the effective

sample size because messages from one writer are dependent.

5.4 Language-only claim gates

All principal gates in this paper are language gates. Acoustic reference duration, speaker similarity, emotion perception, and control leakage are explicitly out of scope.

5.5 Compute and reproducibility

The transparent baseline uses a signed hashed input. The contextual model uses frozen all-MiniLM-L6-v2 representations, a 128-unit projection, and a 96-token limit. The reported PersonaChat run uses one epoch and batch size 128; the matched compositional proxy uses a fixed maximum of 12 epochs and batch size 32, with the selected epoch recorded in each checkpoint. Both use AdamW and gradient clipping at 1.0. Seeds, dataset fingerprints, schema version, epoch, threshold, and metrics are stored with planner checkpoints. This path is designed to run on commodity hardware. Any downstream realizer must record its own hardware, model, and decoder revisions rather than mixing those measurements into this paper.

5.6 Model selection and statistics

Model selection is lexicographic. Configurations must first satisfy a minimum citation precision and maximum false-acceptance rate. Among feasible models we maximize coverage; style accuracy breaks remaining ties. This prevents a weighted average from exchanging a major safety regression for a small style gain. Thresholds may differ by evidence type but not by test speaker.

Paired comparisons use per-speaker differences

Layer	Primary question	Required metric	Failure that blocks a claim
Evidence	Is accepted content attributable?	Citation precision/recall, claim coverage	Missing or wrong evidence ID
Conflict	Is disagreement preserved?	Conflict precision/recall/F1	Contradiction collapsed to consensus
Selection	Does the system know when to stop?	ECE, risk-coverage, false acceptance	High-confidence unsupported acceptance
Style	Is style distinct from topic?	Authorship, style distance, topic controls	Topic-only discrimination

Table 1: Claim gates. Passing one layer does not compensate for failure in another.

and percentile bootstrap intervals over speakers. Binary outcomes also receive a paired permutation test at speaker level. Secondary contrasts use the Holm correction. We report effect sizes and practical margins rather than treating $p < .05$ as evidence of equivalence.

Calibration is assessed globally and by evidence budget, language, and source composition. Because ECE depends on binning, equal-width bins are predeclared; we also report Brier score and reliability diagrams. Risk-coverage curves use every unique acceptance threshold. Their area is a summary, not a replacement for the preregistered operating point.

6 Results and Submission Gate

Table 2 reports the completed public role-instance style proxy under both the original cyclic-negative condition and a harder topic-matched condition. Historical JSON files, UI demonstration values, untrained adapters, and synthetic curves are excluded. PersonaChat measures whether a candidate reply matches the assigned dialogue role given that role’s persona statements; it does not measure a stable natural person, memorial fidelity, factual support, or voice identity.

The full planner’s principal endpoint is citation precision at a declared operating point, not the largest raw coverage. Table 3 reports the frozen SNLI–PersonaChat compositional test over three seeds. A frozen semantic citation prior plus a learned bounded residual raises citation recall while a two-layer selective head ranks jointly valid cases. Selection and component-gate thresholds are separate parameters; an implementation audit found and corrected an earlier contract that accidentally reused the selection threshold for every component gate.

The planner reaches $90.39\% \pm 0.33$ micro citation precision, $74.03\% \pm 0.87$ recall, and $68.70\% \pm$

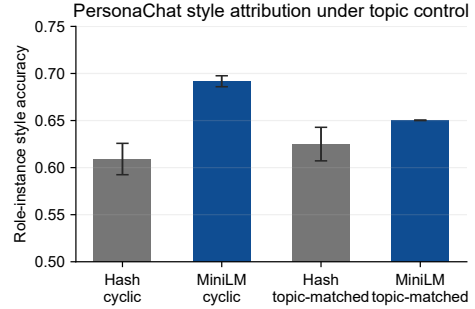


Figure 4: Measured public proxy comparison under cyclic and topic-matched negatives. Error bars show a bootstrap interval for the one-seed hash runs and across-seed standard deviation for MiniLM; they therefore answer different uncertainty questions.

0.77 conflict F1. Positive-case coverage rises from zero in the legacy SciFact diagnostic to $3.73\% \pm 4.24$ here, but remains too low. More importantly, a maximum 5% development false-acceptance constraint yields $23.62\% \pm 3.17$ on test. H1 therefore fails: the development operating point is not a transferable safety guarantee. No seed is selected post hoc.

Table 4 and Figure 5 compare attribution under exactly the same episode-local evidence. Top-1 retrieval is a strong baseline. Selective RAG-2 exchanges recall for precision; only MiniLM RAG-2 meets the 90% test precision level. The planner has comparable precision but lower recall because its citation cutoff is coupled to a broader joint task. At the 1,500 SNLI source-group level, its recall difference from MiniLM RAG-2 is -0.73 points (paired bootstrap 95% CI $[-1.87, 0.43]$; Holm-adjusted sign-flip $p = .210$), so we do not detect a difference for that contrast. It is significantly lower than both top-1 policies and MPNet RAG-2.

6.1 Evidence budget and provenance ablation

The frozen test rows expose a useful distinction between adding evidence and removing the prove-

Condition	Model	Seeds	Acc. / ECE
Cyclic	Hash	1	60.90 [59.25, 62.58] / 3.48
Cyclic	MiniLM	3	69.17 $\pm$ 0.58 / 3.07
Topic	Hash	1	62.53 [60.72, 64.28] / 7.06
Topic	MiniLM	3	65.03 $\pm$ 0.03 / 4.44

Table 2: Measured PersonaChat role-instance style attribution (percent; ECE is reported in percentage points). Accuracy intervals for one-seed rows are speaker bootstrap intervals; multi-seed rows report mean $\pm$ sample standard deviation and average ECE. Topic-matched negatives reduce MiniLM accuracy by 4.14 points relative to cyclic negatives, while preserving a 2.50-point advantage over the matched hash baseline.

Model	Style	Cite P	Cite R	Conflict	Pos. cov.	False acc.	ECE
Selective planner	72.27 $\pm$ 0.44	90.39 $\pm$ 0.33	74.03 $\pm$ 0.87	68.70 $\pm$ 0.77	3.73 $\pm$ 4.24	23.62 $\pm$ 3.17	15.21 $\pm$ 1.89

Table 3: Measured independent-label compositional test (percent; mean $\pm$ sample standard deviation over three seeds). False acceptance is the fraction of accepted cases whose joint target is negative. The development ceiling was 5%; its test violation is a failed hypothesis, not a safety claim.

Equal-budget policy	Cite P	Cite R
MiniLM top-1	88.73	88.73
MiniLM selective RAG-2	91.51	74.77
MPNet top-1	89.63	89.63
MPNet selective RAG-2	89.48	78.00
Planner (3 seeds)	90.39 $\pm$ 0.33	74.03 $\pm$ 0.87

Table 4: Micro citation scores on the SNLI composition test (percent). RAG-2 thresholds are fit only on development data; no method retrieves evidence outside the episode.

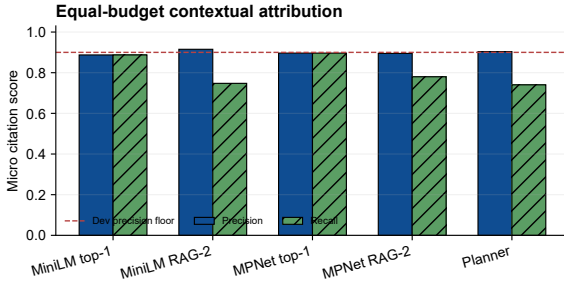


Figure 5: Measured equal-budget contextual baselines. The dashed line is the predeclared 90% development precision floor, not a test guarantee.

nance contract. Figure 7 pools the three registered seeds and groups by the actually observed episode budgets (5, 6, 9, 10, 11; the two larger cells are retained but not interpreted because they contain fewer than 20 rows). The constrained planner accepts only a few percent of positive cases, but its acceptance error is generally lower than a provenance-free replay. The replay uses the fixed score $p_s p_y$ and a threshold of 0.5; it has no citation, conflict, or evidence type gate and is not tuned on the test split. Its positive coverage is roughly 46–54%, accompanied by 34–48% false acceptance.

Budget	Cite P	Cite R	Pos. cov.	False acc.	Status
1	–	–	–	–	not run
2	–	–	–	–	not run
3	–	–	–	–	not run
5	94.12	75.90	3.64	23.53	measured
10	100.00	76.29	2.02	50.00	measured

Table 5: Requested evidence-budget grid on the frozen test snapshot. Cite P is accepted-case citation precision; coverage is positive-case coverage and false acceptance is the fraction of accepted cases with a negative joint target. Dashes are deliberate: no low-budget planner re-evaluation is available.

This is the operational value of typed evidence and conflict preservation: they reduce the set of fluent but unauthorized candidates, even though the present calibration does not achieve the desired coverage.

The requested $\{1, 2, 3, 5, 10\}$ evidence grid is only partially observable in the released predictions. Table 5 reports the two requested cells that are genuinely present; the 1/2/3 cells are marked as unavailable because truncating a 5–11-item prediction would not reproduce a planner inference. This distinction matters: evidence budget is an input to retrieval, attribution, and conflict scoring, not merely a post-hoc mask.

The failure ledger makes the negative result more specific. The source audit contains 17 false acceptances tagged as selection-calibration errors and two as missing support, alongside 1,357 false abstentions at the selection threshold and 87 attribution-underconfidence cases. A secondary replay labels the 17 negative-style accepts as style-overfit and reproduces the two missing-support rows across seeds; no accepted contradictory row

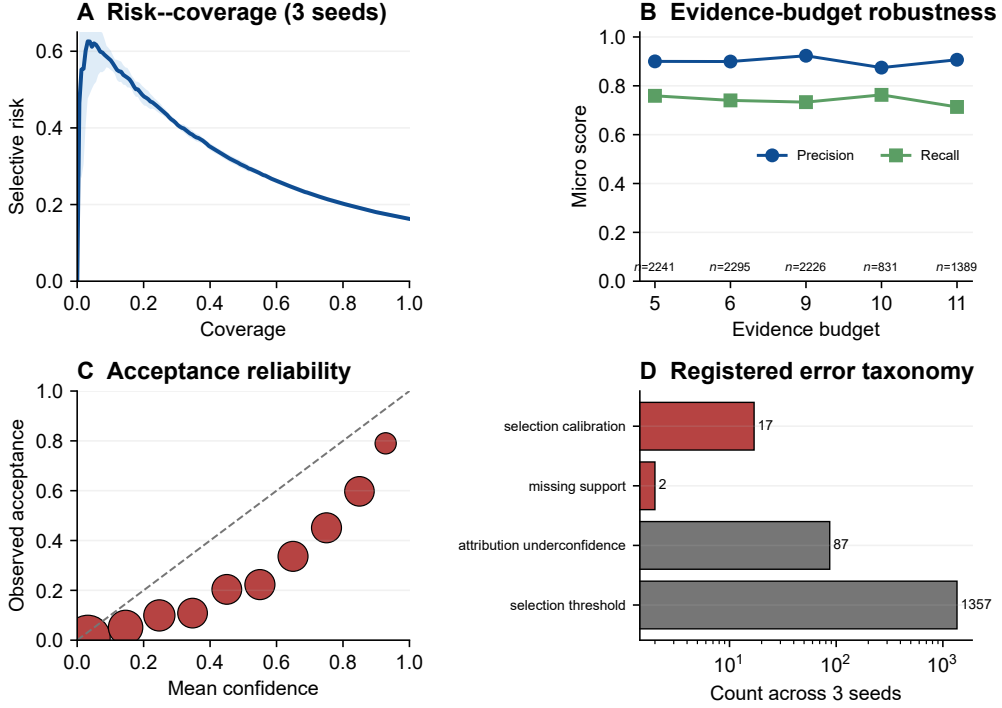


Figure 6: Measured SNLI-composition diagnostics. Shading in A is one sample standard deviation over three seeds. B excludes evidence-budget cells with fewer than 100 pooled observations and prints each retained n . Bubble area in C reflects bin count. D uses a log count axis; red denotes false acceptance and gray false abstention.

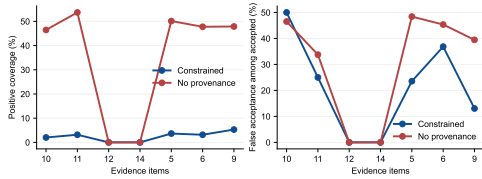


Figure 7: Evidence-budget replay on the untouched test rows. The constrained decision retains provenance and conflict gates; the no-provenance score is the fixed $p_s p_y$ replay. Budgets are the observed cells, not an invented regular grid.

Budget	Sel. abst.	Attr. abst.	Miss. sup.	Calib. FA
5	14.55%	0.80%	0.04%	0.13%
6	15.69%	0.39%	0.00%	0.31%
9	14.60%	1.48%	0.00%	0.13%
10	11.43%	0.24%	0.12%	0.12%
11	18.07%	1.80%	0.00%	0.22%

Table 6: Failure rates normalized by all pooled frozen rows in each observed evidence-budget cell. Selection-threshold false abstention dominates every cell; the accepted-error rates remain sparse but do not vanish with more evidence.

appears in this benchmark. That zero is a construction limitation, not evidence that conflict handling is solved: the composition supplies few adversarial contradictions and should be extended with naturally disputed, source-typed episodes before any deployment claim.

6.2 Cross-domain structural audit

To test whether the evidence-budget interface is tied to PersonaChat’s conversation format, we ran a separate deterministic audit on the official DailyDialog test archive (Li et al., 2017) (958 eligible dialogues and one fixed target turn per dialogue). This auxiliary audit does not rerun the learned planner and

does not introduce entailment annotations. A prior turn is defined as lexical evidence when it shares a non-stopword with the target; the definition is deliberately weak and is reported as such. The result is a useful stress test of evidence availability and citation dilution, not a cross-dataset safety claim.

The monotone increase in lexical support coverage is accompanied by lower top-1 precision as additional turns create more plausible sources. This direction matches the provenance concern exposed by the SNLI composition: more evidence can increase recall while making source attribution less selective. Because DailyDialog has no speaker authorization or claim-level support labels in this release, we do not pool its numbers with the principal

Budget	Eligible	Support cov.	Top-1 P	Top-1 R
1	276	28.81%	100.00%	100.00%
2	418	43.63%	87.32%	100.00%
3	468	48.85%	80.02%	100.00%
5	534	55.74%	70.84%	100.00%

Table 7: DailyDialog lexical-evidence audit on the official test split. The support label is token overlap rather than human entailment; top-1 recall is therefore an availability diagnostic, not factual citation recall.

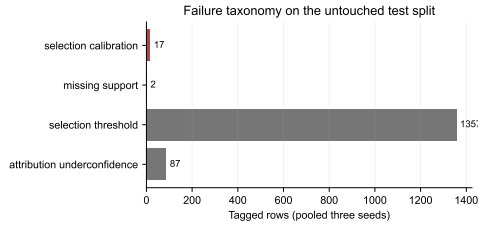


Figure 8: Registered failure tags pooled over three seeds. Selection thresholding dominates false abstention; the accepted-error mass is small but concentrated in calibration/style-overfit cases.

endpoint.

Table 8 preserves the declared architectural ablations on the earlier SciFact–PersonaChat transfer diagnostic. The single-seed rows use the earlier v5 checkpoints loaded through the explicit v7 migration path; they are included as diagnostics, not as additional seed evidence. Removing observable style features reduces style attribution, while removing attention changes the trade-off rather than establishing a universal win. The no-selective row uses the registered legacy support–style–conflict product with its own development calibration.

7 Error Analysis Protocol

Every false acceptance receives one primary and any number of secondary tags: missing support, wrong source, wrong period, unresolved contradiction, quotation inflation, topic–style confusion, calibration error, or policy bypass. False abstentions are divided into retrieval failure, attribution under-confidence, over-sensitive conflict detection, and style under-confidence. Counts and rates are reported by evidence budget and source composition.

Qualitative examples contain synthetic or separately release-authorized text, never private test messages. Each shows typed evidence IDs, gold and predicted citations, probabilities, decision reasons, and the minimal change that flips the decision.

Ablation	Style	Cite P	Cite R	Conflict F1	ECE
Full (3 seeds)	67.64	88.63	35.33	30.41	4.01
No attention	68.88	94.35	27.99	6.94	3.23
No style	63.56	94.96	27.03	1.41	1.44
No selective	66.49	94.31	27.75	29.87	3.40

Table 8: Legacy single-seed ablations on the SciFact compositional test split (percent; ECE in percentage points). Full-row values are the three-seed means; ablations are measured diagnostics with separate development calibration and are not selected on test performance.

Failures are included. Counterfactual examples exchange topic while approximately holding function-word and punctuation profiles fixed, or exchange style while holding topic fixed, probing whether the style head uses content shortcuts.

Audio errors and acoustic leakage are outside this paper’s error analysis. Across three SNLI-composition seeds, the registered taxonomy contains 1,357 positive cases stopped by the selection threshold and 87 stopped by attribution under-confidence. The 19 false acceptances comprise 17 selection calibration errors and two missing-support cases. These counts explain why increasing coverage by simply lowering the threshold is not justified. The released ID-only artifact `snli_error_analysis.json` contains the same tags by evidence budget and capped episode examples; it contains no private message text.

8 Reproducibility Artifacts

The implementation writes immutable JSONL episodes, a manifest fingerprint, planner checkpoints, and machine-readable evaluation rows. A checkpoint contains its schema, dimensions, seed, epoch, threshold, data fingerprint, and measured development metrics. Loading may require an expected fingerprint, preventing accidental evaluation against a different manifest with the same filename. The contextual-baseline artifact records frozen encoder revisions, local-only loading, validation and test fingerprints, top- k , calibrated threshold, and the equal-evidence-budget policy. The three-seed aggregate exposes positive coverage directly rather than requiring it to be inferred from abstention. The paired-analysis artifact averages the two style variants within each of 1,500 SNLI source groups, then records bootstrap intervals, seeded sign-flip tests, and Holm-adjusted p values.

Conceptual figures are deterministic. Result figures reject inputs not marked as measured

evaluation runs and fail when rows are absent. Code, frozen predictions, and SHA-256 manifests for this version are available in the [paper-supplement-v1.0 GitHub snapshot](#). The exact artifact snapshot corresponding to this preprint is tagged `paper-language-v1.0`. The release separates code, consent-safe manifests, model cards, and any restricted data-access process; no private messages or raw audio are redistributed.

9 Ethics, Governance, and Threat Model

The system requires an explicit authorization basis, declared purpose, synthetic-content disclosure, and a revocation channel before synthesis. The server validates these fields before model loading and stores hashes rather than raw submitted text or audio in its audit record. Memorial-mode content also requires a server-validated plan; client-provided “allowed” flags are ignored.

Threats include impersonation, fraud, harassment, reputational harm, fabricated political or financial statements, family conflict, privacy leakage, and emotional dependency. Voice resemblance amplifies these risks. A research prototype therefore uses living-speaker proxy data, visible disclosure, fail-closed consent, exact-quotation fallback, rate limiting at deployment, and revocation. Release should omit private training data and unrestricted identity checkpoints.

Consent from one relative does not settle ownership, privacy, publicity, copyright, or moral rights. Applicable law varies by jurisdiction. A future study with bereaved people would require prospective ethics review, a distress and withdrawal protocol, data-protection review, and careful eligibility rules; approval cannot be obtained retrospectively. No psychological benefit is claimed by the present technical work.

10 Related Work

Style and personality. Language correlates with personality labels at population scale ([Mairesse et al., 2007](#); [Schwartz et al., 2013](#)), but this does not license recovering an individual’s “true personality” from sparse family-selected text. Authorship attribution has a long stylometric tradition ([Stamatatos, 2009](#)), while content-independent representations remain vulnerable to topic effects ([Wegmann et al., 2022](#)). Persona dialogue models condition generation on speaker descriptions or histories ([Li et al., 2016](#); [Zhang et al., 2018](#)); our question is instead

whether a candidate can be authorized by sparse typed evidence without converting persona descriptions into factual claims.

Grounding and selection. Citation-aware generation measures whether claims are attributable ([Gao et al., 2023](#)), while atomic factuality metrics expose errors hidden by document-level scores ([Min et al., 2023](#)). Selective NLP adds the ability to abstain under uncertainty ([Xin et al., 2021](#); [Kamath et al., 2020](#)). Our setting combines these requirements with typed private evidence and explicit source disagreement. Fact-verification benchmarks provide support and contradiction supervision ([Thorne et al., 2018](#); [Wadden et al., 2020](#)), but their labels do not encode speaker consent, source authority, personal style, or authorization to speak in the first person. Retrieval-augmented and self-critical generation ([Lewis et al., 2020](#); [Asai et al., 2024](#); [Manakul et al., 2023](#)) likewise require an explicit evidence contract before use in this setting.

Downstream realization. Zero-shot voice generation can make unsupported language more persuasive ([Chen et al., 2025](#); [Peng et al., 2024](#)). We therefore expose a fail-closed approved-text boundary, but acoustic modeling and controllability are not contributions of this paper.

11 Limitations

The hashed encoder is a transparent baseline, not a competitive contextual model. Speaker-disjoint proxy data cannot reproduce posthumous consent ambiguity, archival degradation, family dynamics, or grief. First-party text may be too sparse to identify stable style; uncertainty should then increase rather than be filled by relatives’ impressions. Observable style features are language- and culture-dependent, and markers require validation beyond English and Mandarin.

Evidence attribution does not prove truth. A first-party message can be false, sarcastic, outdated, quoted from someone else, or outside its consent scope. Atomic claim annotation is costly and may disagree. Calibration can shift across domains, languages, source mixtures, and evidence budgets. It does shift here: a development false-acceptance ceiling of 5% becomes $23.62\% \pm 3.17$ on the independent test split. The current method is therefore unsuitable for a deployment claim even though citation retrieval is strong.

The measured style comparison uses three

contextual-model seeds and one hash seed in each negative policy. PersonaChat roles remain constructed proxies, and the topic-matched policy controls lexical overlap rather than establishing causal topic independence; it therefore motivates, but does not replace, jointly annotated speaker-disjoint data. SNLI relations and PersonaChat style labels were independently collected, but their composition is programmatic. It cannot measure natural candidate generation, pragmatic implicature, long-document retrieval, consent, or a stable person’s style. Strong top-1 retrieval also shows that the benchmark’s attribution component remains substantially semantic and should not be used to claim broad RAG superiority. Finally, an accepted plan can still be harmful. Technical controls cannot resolve who has moral authority to represent a deceased person, and double blind review does not remove these societal questions. Deployment requires jurisdiction-specific review and ongoing revocation, monitoring, and incident response beyond this research prototype.

12 Conclusion

Memorial voice generation should not begin by asking how convincingly a system can imitate a voice. It should first ask what evidence authorizes a candidate utterance, what is merely stylistic resemblance, where sources disagree, and when the system must stop. We provide an executable formulation of this NLP problem, a selective evidence-level planner, and a speaker-disjoint proxy protocol and an independently labelled compositional benchmark. The planner improves citation and conflict measurements, while equal-budget contextual retrieval remains highly competitive. The selective-risk hypothesis fails under split shift, so the present result supports benchmark and diagnostic claims, not safe deployment. The system is not a claim of reconstructing a person, and it makes no acoustic contribution.

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